

Q1. I have doubts on MOSFET. Please explain how a MOSFET works.

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A1. MOSFET/IGFET belongs to the family of FETs. In these FETs, the gate terminal is insulated from the semiconductor body by a very thin layer of silicon dioxide, hence the name Insulated Gate Field Effect Transistor, or IGFET. Also, metal-oxide silicon material is used in their construction, hence the alternative name

of MOSFET.

MOSFET is a three-terminal device: source, gate and drain, composed of a channel of n-type or p-type semiconductor material, and is accordingly called an n-MOSFET or a p-MOSFET. Unlike FET, the gate is separated from the channel by a thin insulating layer of silicon dioxide. The device is available in depletion mode as well as in enhancement mode. Construction and symbol of the device in two modes are shown in Fig. 1 below.

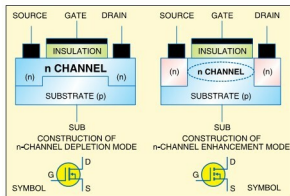


Fig. 1: Construction and symbols of MOSFET

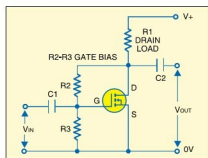


Fig. 2: Linear inverting amplifier circuit

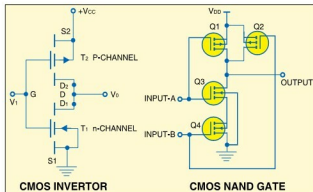


Fig. 3: Digital application circuit using MOSFETs

Working of a MOSFET

Depletion mode. Device has a normally open n-type channel between drain and source. The channel width is controlled by the electrostatic field of the gate bias. The channel can be closed by applying suitable negative bias, or can be increased by applying positive bias. Varying the

voltage between the gate and source modulates the width of the channel, making it possible to control the current flow between drain and source.

Enhancement mode. The conduction channel in the device is normally closed. No n-channel drain to source conduction path exist through the p-type substrate, so with zero bias there is no conduction between drain and source; this feature is indicated in the

symbol of the device also. The channel can be opened by applying forward bias to its gate. Therefore, to turn the device on, significant positive gate bias is needed. Under such condition, p-type substrate material under the gate is converted into a channel, enabling

conduction to take place.

MOSFET application circuits

Brief description of applications of MOSFETs in different modes are as under:

Analogue application. Fig. 2 shows how to use n-channel MOSFET as a linear inverting amplifier. The arrangement is similar to what we have for a CE amplifier using npn transistor. However, operation is a bit different. Gate voltage, not base current, controls the drain current.

Digital application. Major application of MOSFETs is in CMOS where a p-channel and n-channel pair

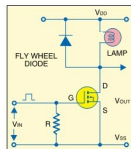


Fig. 4: MOSFET as a switch

of MOSFETs are wired in series but share common input and output terminals. CMOS have applications in analogue as well as in digital circuits.

CMOS logic have advantages, such as very high input impedance, small power consumption, large fan out, speed of operation dependant on the substrate capacitance, and operate with supply voltage ranging from 3 V to 15 V.

MOSFET as a switch. By applying a suitable drive voltage to the gate, the resistance of the drain-source channel can be varied from an 'off-resistance' of many hundreds of k Ω 's, effectively an open circuit, to an 'on-resistance' of less than 10 Ω , effectively a short circuit. This ability to turn the power MOSFET 'on' and 'off' allows the device to be used as a very efficient switch with switching speeds much faster than standard bipolar junction transistors'.

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